InsightIQ

# Dataset & Architecture Migration Report

## ProductID-based procurement → SKU_ID-based procurement

Migration date: 2026-09-27 · Report generated: 2026-09-27 21:34 · Simulation seed: 20260926

Before-state captured 2026-09-27T17:06:35 · after-state captured 2026-09-27T21:22:29 · validation run 2026-09-27T17:23:47

Counts, statistics and test results in this report are read at build time from the migration artefacts in docs/migration/ (state snapshots, validation results, change log), the simulation metadata, and the current and backed-up databases. Explanatory text (reasons, assumptions) is written by the author.

| Headline | Result |
|---|---|
| Raw source data | Unchanged — SHA-256 identical for all 6 raw CSVs; 0 raw records deleted |
| Old behaviour (re-run of pre-migration code) | 53 urgent products → recommended {'SUP_1': 53} · backup {'SUP_9': 53} |
| New behaviour | 11 urgent SKUs · 7 planned with 6 different suppliers · 4 correctly get no supplier (no single supplier has enough) · 50 distinct rankings across 50 SKUs |
| Simulated fields | available_qty, defect_rate, fulfillment_rate (211 real SKU-supplier pairs, seed 20260926) |
| Validation | 27 checks passed · pytest: 82 passed in 12.91s |

# 1. Executive Summary

**The original problem.** Every time procurement was triggered, the system recommended SUP_1 with SUP_9 as backup. Re-running the pre-migration code on the backed-up database confirms it: all 53 urgent products received exactly that pair. This was not a coding error in the ranking itself; the inputs made any other outcome impossible:

- Supplier scores did not depend on the product: one static row per supplier was ranked for every request.
- The eligibility check compared a product's deficit with the supplier's total stock. That stock was actually the buyer's own warehouse inventory (Inventory_Level follows buyer stock flow on 100.0% of rows) and was always larger than any deficit, so no supplier was ever filtered out.
- Supplier quality (rating, defect rate) came from a round-robin assignment of 5 KPI vendors to SUP_1–SUP_10 with no shared key.
- The product table (product.csv) has no relationship to any supplier or SKU: its SupplierID equals ProductID on all 1,000 rows. 26 of its 53 urgent products were discontinued.

**Why SKU_ID.** supply_chain_dataset1.csv is the only dataset where items, suppliers, cost and lead time are genuinely connected: 50 SKUs, 10 suppliers, 211 real SKU-supplier pairs, 3–5 candidate suppliers per SKU.

**What changed.** Procurement requests are now SKU-based and built from the latest real supply-chain snapshot; candidate suppliers are only those that actually supply the SKU; eligibility compares the requested quantity with each supplier's availability; unsupported mappings were removed; the Knowledge Agent's rules were rewired to the new inputs without inventing thresholds.

**What was added.** A real, lossless lane table (sku_supplier_lanes, 250 rows) and a clearly labelled simulated table (supplier_sku_profile, 211 rows) for the three facts the data does not contain.

**What was removed.** The supplier_profiles table and its round-robin mapping, three derived product columns, one index, three config constants, one obsolete function and endpoint. No raw source record was deleted (Section 6).

**What remains simulated.** Supplier available quantity, supplier defect rate and supplier fulfillment rate — each derived from real distributions and documented in Section 8.

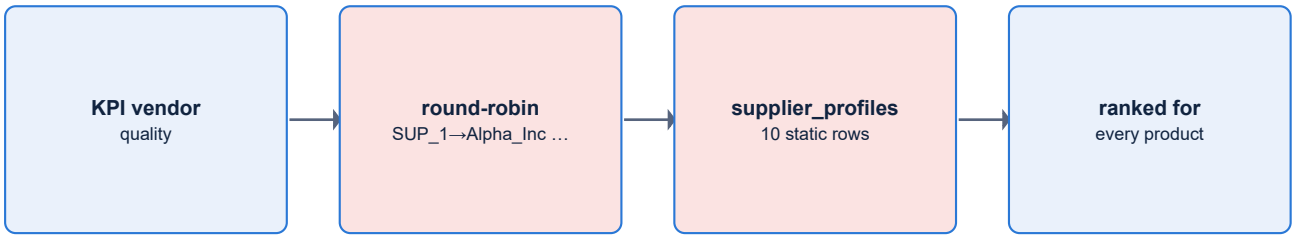
**Future compatibility.** The full 91,250-row daily time series is retained; raw supplier criteria are kept on every candidate; ranking weights are injectable; the agents run on any source that fills the same tables (tested with non-CSV data). Delivery-time ML remains unsupported by the data (Section 14).

# 2. Original Data Architecture

**Original datasets (no shared keys between them)**




**Original relationships**




**Original procurement flow**




## Problems discovered

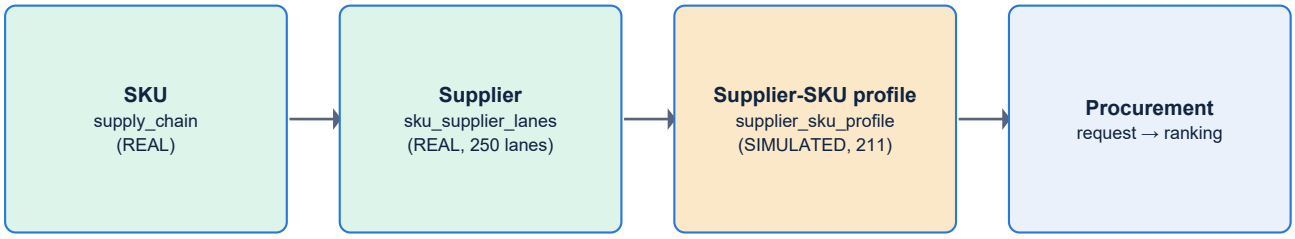
- Disjoint ID spaces: product.csv (1,000 ProductIDs), KPI (5 vendor names), supply_chain (SUP_1–10, SKU_1–50) share no join key.
- Round-robin mapping gave SUP_x the quality history of whichever vendor matched its sorted position.
- stock_availability summed buyer warehouse stock, not supplier stock; smallest value (6,670) exceeded every deficit (max 2,723).
- Product fields are uniformly random and 26 urgent products were discontinued; order.csv has no product reference.
- Old Rule 2 could never fire (max rating 4.39 < 4.5; min delivery 7.9 d > 3 d); Rule 4 fired only for SUP_4 (a relative index maxed at 1.0).

## The deleted round-robin supplier_profiles table (from state_before.json)

| Supplier | Mapped to | Rating | Defect | Reliability | Delivery d | 'Stock' |
|---|---|---|---|---|---|---|
| SUP_1 | Alpha_Inc | 4.39 | 0.0236 | 0.752 | 8.9 | 11867 |
| SUP_2 | Beta_Supplies | 3.9 | 0.0978 | 0.5 | 9.3 | 12604 |
| SUP_3 | Delta_Logistics | 3.6 | 0.1463 | 0.5 | 9.0 | 11167 |
| SUP_4 | Epsilon_Group | 4.37 | 0.0307 | 1.0 | 9.3 | 11498 |
| SUP_5 | Gamma_Co | 4.16 | 0.0502 | 0.872 | 7.9 | 11479 |
| SUP_6 | Alpha_Inc | 4.39 | 0.0236 | 0.758 | 8.7 | 8871 |
| SUP_7 | Beta_Supplies | 3.9 | 0.0978 | 0.5 | 8.7 | 16017 |
| SUP_8 | Delta_Logistics | 3.6 | 0.1463 | 0.5 | 9.2 | 11297 |
| SUP_9 | Epsilon_Group | 4.37 | 0.0307 | 0.597 | 9.0 | 6670 |
| SUP_10 | Gamma_Co | 4.16 | 0.0502 | 0.572 | 8.2 | 12549 |

# 3. New Data Architecture

**New relationships**




- SKU identity and urgency: supply_chain (per SKU × warehouse × day). A SKU is urgent if any warehouse lane is below its reorder point on the latest date (2024-12-30).
- Supplier relationship: sku_supplier_lanes — each (SKU, warehouse) has exactly one supplier with a real unit cost and lead time that never change during the year (lossless).
- Supplier-SKU profile: supplier_sku_profile keyed by the 211 real (SKU, supplier) pairs; adds simulated availability and supplier-level quality.
- Candidate matrix for a request = suppliers serving the SKU (lanes) joined with their profile row. Where a supplier serves the SKU at 2–3 warehouses (36 pairs), cost and lead time are averaged and min/max/lane count are kept.
- required_quantity = sum over urgent lanes of the lane's median historical order size, rounded up to 10 — replicating the real (s, Q) policy: orders occur only below the reorder point, and lot size is unrelated to the shortfall.

# 4. Dataset-by-Dataset Changes

## Files (raw and generated)

| File | New? | Rows before | Rows after | Cols before | Cols after | Status |
|---|---|---|---|---|---|---|
| Procurement KPI Analysis Dataset.csv | - | 777 | 777 | 11 | 11 | unchanged (hash identical) |
| archive (1)/clander.csv | - | 1000 | 1000 | 10 | 10 | unchanged (hash identical) |
| archive (1)/customer.csv | - | 1000 | 1000 | 10 | 10 | unchanged (hash identical) |
| archive (1)/order.csv | - | 1000 | 1000 | 13 | 13 | unchanged (hash identical) |
| archive (1)/product.csv | - | 1000 | 1000 | 9 | 9 | unchanged (hash identical) |
| simulated/evidence/order_quantity_by_pair.csv | new file | - | 211 | - | 15 | ADDED |
| simulated/supplier_sku_profile.csv | new file | - | 211 | - | 5 | ADDED |
| supply_chain_dataset1.csv | - | 91250 | 91250 | 15 | 15 | unchanged (hash identical) |

No raw file was renamed, and no raw column was added, removed, renamed or transformed. Transformations happen only when loading into SQLite (below).

## SQLite tables

| Table | Rows before | Rows after | Cols before | Cols after | Columns added | Columns removed | Reason |
|---|---|---|---|---|---|---|---|
| calendar | 935 | 935 | 9 | 9 | | | Pre-existing: 65 duplicate dates filtered (unchanged) |
| customers | 1000 | 1000 | 9 | 9 | | | unchanged |
| orders | 1000 | 1000 | 11 | 11 | | | unchanged |
| procurement_kpi | 777 | 777 | 16 | 16 | | | unchanged |
| products | 1000 | 1000 | 12 | 9 | | deficit, target_stock, urgent | Derived procurement columns removed; reference only |
| purchase_orders | 0 | 0 | 13 | 14 | sku_id | | sku_id added by ALTER (rows preserved) |
| sku_supplier_lanes | - | 250 | - | 5 | table added | - | Real per-lane supplier cost/lead time |
| supplier_profiles | 10 | - | 13 | - | - | table deleted | Unsupported round-robin mapping; mislabelled stock |
| supplier_sku_profile | - | 211 | - | 5 | table added | - | Simulated availability + quality |
| supply_chain | 91250 | 91250 | 15 | 15 | | | unchanged |

Rows: 0 rows deleted from any table carried over. Rows added: 250 (lanes) + 211 (profile). Rows filtered: 65 duplicate calendar dates (pre-existing behaviour); all 1,000 products excluded from procurement but retained. Renamed: config weight keys w1_rating → w1_quality, w4_reliability → w4_fulfillment; proposed field on_time_rate → fulfillment_rate. Transformed: products.reorder_level clip(lower=1) removed; SUP_3 defect rate capped.

# 5. Detailed Change Table

28 entries. Full machine-readable version with evidence, method and impact: docs/migration/change_log.json.

| Dataset / location | Field | Change | Before | After | Reason | Status |
|---|---|---|---|---|---|---|
| All 6 raw CSVs: DataSets/ | all | RETAINED | 6 files | 6 files, SHA-256 identical | Raw data must stay recoverable and unchanged | REAL |
| Raw checksum manifest: DataSets/CHECKSUMS.sha256 | - | ADDED | - | 6 SHA-256 lines | Proves raw files are unchanged | N/A |
| Backup: data/backup/pre_migration_2026-09-27/ | - | ADDED | - | 6 raw CSVs + pre-migration insightiq.db | Original data must remain recoverable | REAL |
| Simulated supplier-SKU profile: DataSets/simulated/supplier_sku_profile.csv | file | ADDED | - | 211 rows × 5 columns | Fill genuine data gaps (supplier availability and quality) | SIMULATED |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | sku_id, supplier_id | ADDED | - | 211 pairs | Row key; relationships must be real | REAL |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | available_qty | ADDED | - | range 200-499, mean 351.7 | Supplier-side availability does not exist in the source | SIMULATED |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | defect_rate | ADDED | - | Beta(1.5927, 21.8594) draws, cap 0.1442 | KPI vendors cannot be mapped to SUP_x | SIMULATED |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | defect_rate (SUP_3) | TRANSFORMED | 0.1634 | 0.1442 | Approved cap at highest real supplier rate | SIMULATED |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | fulfillment_rate | ADDED | - | Binomial(137, 0.804598)/137 | Needed by Rule 4; no SUP_x performance data exists | SIMULATED |
| Simulated supplier-SKU profile: supplier_sku_profile.csv | on_time_rate → fulfillment_rate | RENAMED | on_time_rate (proposed) | fulfillment_rate | KPI data has no promised delivery date | SIMULATED |
| Simulation metadata: DataSets/simulated/*.meta.json, evidence/, README.md | - | ADDED | - | seed, parameters, input/output hashes, per-pair statistics, data dictionary | Reproducibility and audit | N/A |
| SQLite: supplier_profiles | table | DELETED | 10 rows × 13 columns | - | Built on an unsupported round-robin KPI-vendor→SUP_x mapping | N/A (derived) |
| SQLite: supplier_profiles | benchmarked_against | DELETED | 10 assignments | - | Round-robin mapping with no evidence | N/A |
| SQLite: supplier_profiles | stock_availability | DELETED | sum of latest Inventory_Level by supplier | - | Mislabelled: it was the buyer's own warehouse stock | N/A |
| SQLite: supplier_profiles | rating, reliability_score, delivery_time_days, region | DELETED | KPI rating via mapping; min-max lead-time std index; 0.7/0.3 blend with KPI days; region mode | - | Derived from the invalid mapping or from noise (Region changes daily within a fixed lane) | N/A |
| SQLite: sku_supplier_lanes | table | ADDED | - | 250 rows × 5 columns | Real per-SKU, per-warehouse supplier cost and lead time | REAL (derived) |
| SQLite: supplier_sku_profile | table | ADDED | - | 211 rows × 5 columns | Load of the simulated CSV | SIMULATED (+ real keys) |
| SQLite: supply_chain | all | RETAINED | 91250 rows × 15 | 91250 rows × 15 | Time series needed for future ML/GA | REAL |
| SQLite: products | target_stock, deficit, urgent | DELETED | 12 columns | 9 columns | Only existed for product-based procurement | N/A (derived) |

| Dataset / location | Field | Change | Before | After | Reason | Status |
|---|---|---|---|---|---|---|
| SQLite: products | reorder_level | TRANSFORMED | clip(lower=1) | raw value | Clip only served the urgency calculation | REAL |
| SQLite: products | all rows | FILTERED | 1000 rows eligible for procurement | 0 rows eligible for procurement (table retained) | No ProductID→SKU or ProductID→supplier relationship | REAL |
| SQLite: idx_products_urgent | index | DELETED | index on products(urgent) | - | Column removed | N/A |
| SQLite: purchase_orders | sku_id | ADDED | 13 columns | 14 columns | Procurement identity is SKU | N/A |
| SQLite: indexes | supply_chain(date), supply_chain(sku_id,date), unique lanes, unique profile | ADDED | - | 4 indexes | Query speed and integrity constraints | N/A |
| SQLite: PRAGMA user_version | - | MODIFIED | 0 | 2 | Old databases rebuild automatically | N/A |
| SQLite: calendar | rows | FILTERED | 1000 raw rows | 935 rows | Pre-existing loader behaviour (unchanged by this migration) | REAL |
| Config: backend/config.py | TARGET_STOCK_MULTIPLIER, RULE_PREFERRED_RATING, DEFAULT_DAILY_DEMAND_FALLBACK | DELETED | 3, 4.5, 15 | - | Their only consumers were removed | N/A |
| Config: backend/config.py | w1_rating, w4_reliability | RENAMED | w1_rating 0.30, w4_reliability 0.25 | w1_quality 0.30, w4_fulfillment 0.25 | Inputs changed; values kept | N/A |

# 6. Deletion Report

**No raw source records were deleted.** All 6 raw CSVs have identical SHA-256 before and after, and git reports no change against HEAD. supply_chain keeps all 91,250 rows and 15 columns. Every deletion below is of a derived database artefact, code or behaviour.

| What | Count | Why | Evidence | Future impact | Backup | Replaced by |
|---|---|---|---|---|---|---|
| 1. SQLite table supplier_profiles | 10 rows × 13 columns | Built on an unsupported mapping and a mislabelled stock field | No shared key between KPI vendors and SUP_x; Inventory_Level is buyer stock | None: every real input remains in supply_chain | Yes: data/backup/pre_migration_2026-09-27/insightiq.db and full contents in state_before.json | sku_supplier_lanes (real) + supplier_sku_profile (simulated) |
| 2. Round-robin supplier mapping | 10 assignments: SUP_1→Alpha_Inc, SUP_2→Beta_Supplies, SUP_3→Delta_Logistics, SUP_4→Epsilon_Group, SUP_5→Gamma_Co, SUP_6→Alpha_Inc, SUP_7→Beta_Supplies, SUP_8→Delta_Logistics, SUP_9→Epsilon_Group, SUP_10→Gamma_Co | Relationship with no evidence | Assigned by sorted position only | None | Yes (state_before.json) | Explicitly simulated supplier-level quality |
| 3. Derived products columns target_stock, deficit, urgent | 1000 rows × 3 columns | Only used for product-based procurement | 53 urgent flags, 26 on discontinued products | None: recomputable from raw product.csv | Yes (backup DB); raw CSV untouched | SKU urgency from supply_chain |
| 4. Index idx_products_urgent | 1 index | Indexed a removed column | - | None | Yes (backup DB) | - |
| 5. Products excluded from procurement (FILTERED, not deleted) | 1000 rows | No product→SKU/supplier relationship | SupplierID == ProductID on every row | None: table retained for reference | Rows are still in the database and raw CSV | SKU-based requests |
| 6. Calendar duplicate dates (pre-existing, unchanged) | 65 rows | Loader keeps one row per date | Duplicate parsed dates | None | Raw CSV untouched | - |
| 7. Config constants | 3 constants | No remaining consumers | - | None | git history | See change log |
| 8. Code: perception demand_signals (orders + customers aggregation) | 1 function | E-commerce orders have no product/SKU reference; unused by the UI | order.csv has no product column | None: orders/customers tables retained | git history | Real Units_Sold demand per SKU |
| 9. Behaviour: split-order fallback that ranked the full pool when nobody was eligible | 1 code path | It recommended a supplier that could not fulfil the order | Required by the 'no false selection' requirement | Split-order optimisation can be added later; combined availability is reported | git history | no_eligible_supplier result, no recommendation, no plan |
| 10. API GET /api/products | 1 endpoint | Product table no longer drives procurement | - | None | git history | GET /api/skus (old path returns 410) |

# 7. Simulated Dataset

Table **supplier_sku_profile** · file DataSets/simulated/supplier_sku_profile.csv · 211 rows · 211 valid SKU-supplier relationships (all from the source, 0 invented, 0 missing) · 50 SKUs · 10 suppliers · suppliers per SKU {'3': 7, '4': 25, '5': 18}.

| Field | Type | Status | Granularity | Source / method |
|---|---|---|---|---|
| sku_id | text | REAL | row key | supply_chain SKU_ID (existing relationship) |
| supplier_id | text | REAL | row key | supply_chain Supplier_ID (existing relationship) |
| available_qty | integer | SIMULATED | SKU-supplier | one draw from the pair's own real Order_Quantity history |
| defect_rate | fraction | SIMULATED | supplier | Beta fit to real KPI supplier defect rates, capped at real max |
| fulfillment_rate | fraction | SIMULATED | supplier | Binomial sampling around the real pooled KPI fulfillment rate |

Warehouse is not a column: each (SKU, warehouse) has exactly one supplier in the source, and procurement requests carry no warehouse.

## Supplier-level simulated quality (and availability range across that supplier's SKUs)

| Supplier | defect_rate | fulfillment_rate | Rule 3 High Risk | SKUs | available_qty range |
|---|---|---|---|---|---|
| SUP_1 | 0.0118 | 0.8394 | no | 23 | 200–478 |
| SUP_2 | 0.0833 | 0.8467 | yes | 21 | 228–499 |
| SUP_3 | 0.1442 | 0.8613 | yes | 20 | 207–497 |
| SUP_4 | 0.0194 | 0.7810 | no | 22 | 202–497 |
| SUP_5 | 0.1333 | 0.7737 | yes | 21 | 210–495 |
| SUP_6 | 0.0365 | 0.7664 | no | 16 | 203–479 |
| SUP_7 | 0.0513 | 0.8102 | yes | 29 | 202–498 |
| SUP_8 | 0.0596 | 0.7518 | yes | 22 | 201–478 |
| SUP_9 | 0.0448 | 0.8686 | no | 16 | 222–486 |
| SUP_10 | 0.1365 | 0.8394 | yes | 21 | 236–495 |

## First 10 rows

| sku_id | supplier_id | available_qty | defect_rate | fulfillment_rate |
|---|---|---|---|---|
| SKU_1 | SUP_2 | 457 | 0.0833 | 0.8467 |
| SKU_1 | SUP_5 | 338 | 0.1333 | 0.7737 |
| SKU_1 | SUP_8 | 317 | 0.0596 | 0.7518 |
| SKU_1 | SUP_9 | 418 | 0.0448 | 0.8686 |
| SKU_2 | SUP_2 | 395 | 0.0833 | 0.8467 |
| SKU_2 | SUP_5 | 403 | 0.1333 | 0.7737 |
| SKU_2 | SUP_7 | 311 | 0.0513 | 0.8102 |
| SKU_2 | SUP_9 | 416 | 0.0448 | 0.8686 |
| SKU_2 | SUP_10 | 359 | 0.1365 | 0.8394 |
| SKU_3 | SUP_3 | 317 | 0.1442 | 0.8613 |

# 8. Simulation Methodology

Seed 20260926, fixed before any result was seen and not re-rolled. numpy SeedSequence(seed).spawn(3) gives one independent stream per field, so changing one field's method never shifts another field's values.

## 8.1 available_qty

- Why simulated: supplier-side stock is not in any file; Inventory_Level satisfies inv[t] = inv[t-1] − sold[t] + order_qty[t] on 100.0% of rows, i.e. it is buyer stock.
- Historical source: 5,027 real orders (Order_Quantity > 0); 18–62 per pair (minimum required 10).
- Statistical basis: order sizes 200–499, mean 349.8, median 351, sd 86.2, p10 231, p90 468; decile counts [486, 510, 492, 533, 469, 513, 492, 540, 502, 490] (uniform); pairs do not differ (Kruskal-Wallis p = 0.8672).
- Sampling method: one value drawn uniformly (with replacement) from that pair's own order-size history.
- Validation: every value is a real order size of its pair (True); generated mean/sd 351.66/86.39 vs history 349.83/86.19.
- Limitation: because pairs do not really differ, differences between suppliers come from the draw.

| Request size | Pairs eligible | SKUs with no eligible supplier | SKUs with all eligible |
|---|---|---|---|
| 276 | 76.3% | 0 | 17 |
| 351 | 48.8% | 4 | 3 |
| 425 | 27.0% | 13 | 1 |
| 468 | 10.4% | 30 | 0 |

## 8.2 defect_rate

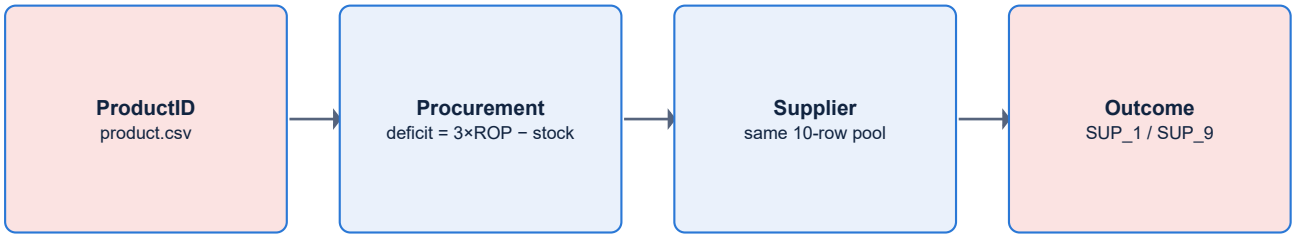
- Why the KPI names cannot map to SUP_x: no shared key, no SKU, no category in supply_chain, and non-overlapping dates (KPI 2022-01 → 2024-01, supply chain 2024).
- Calculation: Defective_Units / Quantity per PO; 136 POs with blank Defective_Units excluded (unknown ≠ zero), 641 used. Supplier rate = Σ defective / Σ quantity: Alpha_Inc 0.0225, Beta_Supplies 0.0935, Delta_Logistics 0.1443, Epsilon_Group 0.0297, Gamma_Co 0.0496.
- Per-PO distribution: min 0.0000, median 0.0515, mean 0.0701, sd 0.0501, p95 0.1576, max 0.3571. Defect rate is a supplier trait (Kruskal-Wallis by supplier p = 2.5e-118), so it is simulated per supplier, never per SKU.
- Distribution: Beta(α = 1.5927, β = 21.8594), method-of-moments fit to the 5 supplier rates (mean 0.0679, sd 0.0509).
- Cap: real maximum 0.1442986 floored to 4 dp = 0.1442 (rounding up would exceed the real max). Capped: SUP_3 0.1634 → 0.1442.
- Result: 6 of 10 suppliers exceed Rule 3's 0.05 (SUP_2, SUP_3, SUP_5, SUP_7, SUP_8, SUP_10) vs 2 of 5 real vendors.

## 8.3 fulfillment_rate

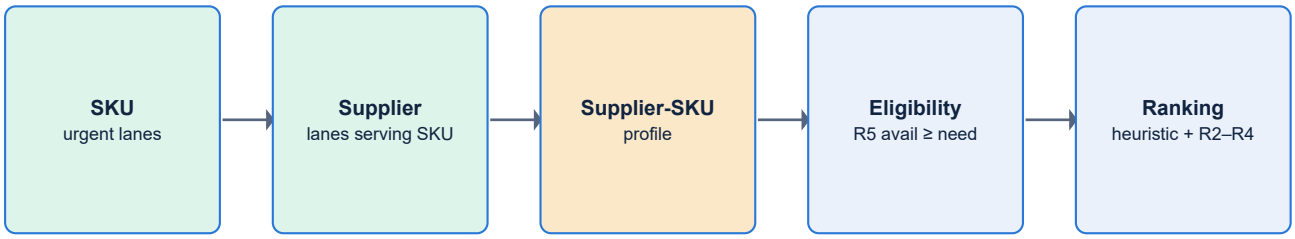
- Why not an on-time rate: the KPI file has Order_Date, Delivery_Date and Order_Status but no promised/expected date, so lateness cannot be measured.
- Definition: Delivered / (Delivered + Partially Delivered + Cancelled); Pending excluded because its outcome is unknown.
- Real values: pooled 0.804598; per vendor Alpha_Inc 0.823, Beta_Supplies 0.803, Delta_Logistics 0.779, Epsilon_Group 0.811, Gamma_Co 0.811; vendors do not differ (chi-square = 0.985, p = 0.912).
- Method: every simulated supplier has the same true rate p = 0.804598; observed rate = Binomial(137, p) / 137 (137 = median finished POs per real vendor).
- Limitation: differences between simulated suppliers are sampling noise by design, and their spread is wider than the real one. The ranking therefore uses the rate absolutely rather than Min-Max stretching it.

# 9. Before / After Architecture

BEFORE




AFTER




| Aspect | Before | After |
|---|---|---|
| Item identity | ProductID (1,000 random products) | SKU_ID (50 real SKUs) |
| Urgency | UnitsInStock < ReorderLevel (product.csv) | Inventory < ReorderPoint per warehouse lane, latest date |
| Quantity | 3 × ReorderLevel − stock | Σ median historical lot size of urgent lanes, rounded to 10 |
| Candidates | All 10 suppliers, every time | Only suppliers that supply the SKU (3–5) |
| Eligibility | Buyer stock ≥ deficit (never failed) | Supplier available_qty ≥ required quantity |
| Nobody eligible | Ranked the full pool and recommended anyway | No supplier selected; combined availability reported |
| Quality source | Round-robin KPI vendor | Explicitly simulated supplier-level profile |
| Result (53 / 11 urgent items) | {'SUP_1': 53} | 7 planned via SUP_2, SUP_3, SUP_4, SUP_6, SUP_7, SUP_9; 4 no supplier |

# 10. Code Changes

Line counts from git diff --numstat against the pre-migration commit.

| File | Lines | What changed / why | Old behaviour | New behaviour |
|---|---|---|---|---|
| backend/agents/coordinator_agent.py | +11 / −8 | Passes sku_id and required quantity | product_id, deficit | sku_id, required_quantity |
| backend/agents/knowledge_agent.py | +98 / −44 | Rules on new fields; R5 capacity; no-eligible decision | Rating/reliability rules | R2 lead+defect, R4 fulfillment ≥ 0.90, R5 capacity |
| backend/agents/perception_agent.py | +107 / −80 | SKU urgency from latest supply_chain date; lanes kept | Read products table | Per-lane Rule 1 inputs, required qty = median lot size sum |
| backend/agents/planning_agent.py | +40 / −27 | No plan when nothing needed / no supplier; real demand | Planned any product, ROP/14 demand | Order = required qty; demand = 28-day mean Units_Sold |
| backend/agents/supplier_agent.py | +142 / −136 | Candidate matrix per SKU, eligibility, injectable weights | Supplier-wide pool, total stock filter, full-pool fallback | Only suppliers serving the SKU; ineligible reported, never ranked |
| backend/config.py | +24 / −9 | Paths, provenance, seed, schema version; renamed weight keys | Product-era constants | SKU-era constants; weights unchanged |
| backend/database.py | +104 / −116 | Removed round-robin profiles; added lanes + validated profile loader; schema v2 migration | Built supplier_profiles; product urgency | Real lanes, simulated profile, auto-rebuild, PO migration |
| backend/main.py | +100 / −72 | SKU endpoints; validation on approve-order | product endpoints | /api/skus, sku_id params, 410 for product_id |
| backend/tests/test_supplier_agent.py | +167 / −178 | Rewritten for candidate matrix and new components | Old schema | 35 tests |
| frontend/static/css/styles.css | +2 / −0 | Ineligible and SIM badges | - | 2 classes |
| frontend/static/js/api.js | +8 / −4 | skus(), sku_id params | products(), product_id | skus(), sku_id |
| frontend/static/js/app.js | +162 / −93 | SKU picker, alert table, matrix, SIM labels, no-eligible states | Product fields | SKU fields |
| requirements.txt | +1 / −0 | + httpx (FastAPI TestClient) | - | httpx==0.28.1 |
| backend/simulation/supplier_sku_profile.py | new | Simulation generator | - | new file |
| backend/tests/test_simulation.py | new | 14 simulation tests | - | new file |
| backend/tests/test_sku_pipeline.py | new | 33 pipeline/API/migration tests | - | new file |
| scripts/migration/*.py | new | State capture, validation, logs, PDF report | - | new file |

## Knowledge Agent rule changes

| Rule | Before | After | Threshold | Justification |
|---|---|---|---|---|
| R1 Inventory state | IF Inventory < ReorderPoint (product) | Same, per urgent warehouse lane | unchanged | Urgency is per lane in the real data |
| R2 Preferred | IF Rating >= 4.5 AND DeliveryTime <= 3d | IF LeadTime <= 3d AND DefectRate <= 0.05 | no new numbers (3 d and 0.05 already existed) | Rating only existed through the removed mapping; old rule could never fire (max rating 4.39, min delivery 7.9 d) |
| R3 High Risk | IF DefectRate > 0.05 | Same (simulated supplier defect rate) | unchanged | - |
| R4 Reliable | IF no status AND Reliability >= 0.90 | IF no status AND FulfillmentRate >= 0.90 | 0.90 unchanged (not lowered) | Old input was a relative min-max index; 0.90 is kept as a 90% fulfillment target. No supplier meets it (real best 82%); lowering it would classify sampling noise, since real suppliers do not differ |
| R5 Capacity (new) | - | IF AvailableQty >= RequiredQty ELSE Ineligible | required quantity | Makes every exclusion visible in the XAI trace |

Heuristic scoring: weights unchanged (0.30 / 0.25 / 0.20 / 0.25). Quality = 1 − defect_rate and Fulfillment = fulfillment_rate are used absolutely (the precedent of the old Rating/5 term); Cost and Delivery keep Min-Max normalization over eligible suppliers. Weights can now be passed in as data (rank_suppliers(..., weights=...)).

# 11. Database Changes

| Item | Before | After |
|---|---|---|
| Tables added | - | sku_supplier_lanes, supplier_sku_profile |
| Tables removed | supplier_profiles | - |
| Columns added | - | purchase_orders.sku_id |
| Columns removed | products.target_stock, deficit, urgent | - |
| Columns renamed | - | none in DB (config keys only) |
| Indexes | idx_calendar_date, idx_products_urgent | idx_calendar_date, idx_supply_chain_date, idx_supply_chain_sku_date, ux_lanes_sku_warehouse, ux_profile_sku_supplier |
| Constraints | none | UNIQUE(sku_id, warehouse_id) on lanes; UNIQUE(sku_id, supplier_id) on profile |
| Schema version (PRAGMA user_version) | 0 | 2 |

## Migration and initialization

- On startup, build_database() compares PRAGMA user_version with SCHEMA_VERSION = 2; an older database is rebuilt from the CSVs automatically.
- Rebuild replaces derived tables, drops obsolete supplier_profiles / idx_products_urgent, and migrates purchase_orders in place with ALTER TABLE ADD COLUMN sku_id — existing purchase orders are kept (tested with a v1 database containing a PO row).
- The profile loader rejects invented (SKU, supplier) pairs, duplicates and out-of-range values; the lane builder fails if a lane's supplier, cost or lead time ever varies.
- At migration time purchase_orders had 0 rows.

# 12. API Changes

| Endpoint | Before | After |
|---|---|---|
| POST /api/run-agent-pipeline | product_id (int, required) | sku_id (str); product_id → 410 Gone |
| GET /api/products | product list | 410 Gone |
| GET /api/skus | - | new: 50 SKUs with urgency, required qty, candidate count |
| GET /api/suppliers | required_deficit | sku_id (required), required_quantity (default: SKU's current need); ineligible list, provenance, weights |
| POST /api/approve-order | product_id | sku_id; 422 if pair not real or quantity > available_qty |
| GET /api/dashboard | product alerts, e-commerce lead time | SKU alerts, snapshot date, real lead time, simulated fulfillment (labelled) |
| GET /api/system-health | products.urgent count | urgent SKU count; table list from schema |

Backward compatibility: requests using product_id receive 410 Gone with an explanation instead of a silent 422, so old clients fail loudly. approve-order now validates the (SKU, supplier) relationship and quantity ≤ available_qty (422 otherwise).

# 13. Frontend Changes

- Dashboard: KPIs are urgent SKUs, real average supplier lead time and simulated average fulfillment (labelled); chart of top SKUs by required quantity; alert table shows SKU, urgent warehouses, inventory, reorder point, shortfall, required quantity, candidate supplier count.
- Pipeline & XAI: 50-SKU picker (urgent SKUs marked with their need); perceived state shows snapshot date, urgent warehouses, required quantity; recommended supplier shows unit cost, lead time, available quantity [SIM] and score; a 'No supplier selected' state explains combined availability and that split orders are not implemented; the trace shows Rule 1 per lane, Rule 5 capacity and an 'Ineligible' badge.
- Supplier Matrix: SKU-scoped with a required-quantity input; formula text built from the weights actually used; columns for unit cost, lead time, available [SIM], defect [SIM], fulfillment [SIM]; multi-warehouse costs marked with * and a min–max tooltip; separate table of excluded suppliers with shortfall.
- Action Hub: plan titled SKU → supplier; approval payload uses sku_id; order history shows SKU.
- Simulated-data labels: an amber SIM tag on every simulated value, with a tooltip pointing to the data dictionary.

- Verified in the running app (all four views, SKU_2 planned, SKU_19 no-eligible, required-quantity re-ranking, no console errors, all API calls 200).

## 14. Future Compatibility

| Future feature | Compatible? | Why / evidence | Limitation |
|---|---|---|---|
| Demand ML | Yes | All 91,250 daily rows × 15 columns retained (365 days per SKU-warehouse, tested); SKU identity unchanged. | Demand_Forecast correlates 0.949 with Units_Sold — keep as baseline, not a feature (leakage). |
| Demand-spike prediction | Yes | Promotion_Flag, warehouse, region, inventory, date retained; requests keep lanes. | Region varies randomly within fixed lanes; SKUs have near-identical mean demand. |
| Delivery-time ML | No | Lead time is constant for every lane (0 lanes vary). lead_time_days is a candidate column, so a model could supply it later without redesign. | Needs real promised vs actual delivery dates per supplier/SKU/order. |
| GA dynamic weighting | Yes (hook) | Weights injectable (tested); delivery-heavy weights change the top supplier for 17 of 50 SKUs; global state available: 0–35 lanes below ROP per day. | No outcome data (stockout flag sum = 0); fitness must come from a disclosed simulation. |
| Split-order optimization | Yes (data) | Per-supplier available_qty kept; combined availability reported; 4 of 11 urgent SKUs need it today. | Allocation algorithm not implemented. |
| AHP | Yes | Candidate matrix keeps raw cost, lead time, availability, defect, fulfillment on every candidate (tested). | Pairwise judgements needed. |
| TOPSIS | Yes | score_candidates / candidate_matrix work on a plain DataFrame (tested). | - |
| ERP integration | Yes | Pipeline tested end-to-end on tables filled from plain records (no CSV); loader validates relationships; simulated columns replaceable. | No live connector; SQLite store. |

Architectural checks from validation: all passed — time-series columns for demand ML retained; 365 daily observations per SKU-warehouse retained; raw criteria on every candidate; per-supplier availability retained (split orders).

## 15. Assumptions

- SKU_ID is the procurement item identity; product.csv takes no part in procurement.
- Procurement requests have no warehouse dimension; candidate suppliers are considered at SKU level.
- A supplier that delivers an SKU to one warehouse could deliver it to any warehouse.
- Where a supplier serves an SKU at several warehouses (36 pairs), its cost and lead time for the SKU are the arithmetic mean of its lanes.
- The latest date in the data (2024-12-30) is 'today' for stock state; planning dates are computed from the real current date.
- Required quantity replicates the observed policy: median historical lot size per urgent lane, summed, rounded up to 10.
- Daily demand = mean real Units_Sold over the last 28 days of the focus lanes.
- Supplier availability is simulated from that pair's historical order sizes.
- Supplier quality (defect, fulfillment) is simulated at supplier level.
- Fulfillment rate is not an on-time rate.
- Rule 4's 0.90 is read as a 90% fulfillment target (unchanged number).
- Future delivery-time ML requires promised and actual delivery dates.

## 16. Limitations

- All three supplier-side fields are simulated; rankings and eligibility depend on them and must not be presented as observed supplier behaviour.
- The underlying source datasets are themselves synthetic (perfect 50 × 5 × 365 grid, uniform order sizes, same-day receipt of orders).

- Defect distribution is fitted to only five real vendors; order sizes and fulfillment do not differ between real entities, so simulated differences there are sampling noise.
- With this seed 6 of 10 suppliers exceed the 0.05 defect threshold; for 3 of 7 planned SKUs the only eligible supplier is High Risk (existing fallback recommends it with an explicit warning).
- No supplier reaches fulfillment 0.90, so Rule 4 never labels a supplier Reliable (none).
- 4 of 11 urgent SKUs cannot be served by one supplier; split-order allocation is not implemented.
- No real on-time, delivery-time or supplier-capacity data exists; delivery-time ML is not supported.
- Calendar seasonality in planning still uses clander.csv, an unrelated practice dataset (unchanged by this migration).
- The backup in data/backup/pre_migration_2026-09-27/ is local only (data/ is git-ignored); git history also holds the raw CSVs.
- Real data required to remove the simulation: supplier available quantity per SKU, supplier defect history for SUP_x, promised vs actual delivery dates.

# 17. Validation Results

Run 2026-09-27T17:23:47 · all checks passed: True · pytest: 82 passed in 12.91s (test_simulation.py: 14, test_sku_pipeline.py: 33, test_supplier_agent.py: 35).

| Group | Check | Result | Detail |
|---|---|---|---|
| Raw data | raw unchanged: Procurement KPI Analysis Dataset.csv | PASS | bf9a529c52cd7de2 |
| Raw data | raw unchanged: archive (1)/clander.csv | PASS | 085b9af1967e7598 |
| Raw data | raw unchanged: archive (1)/customer.csv | PASS | 99ccca120ca9cbcd |
| Raw data | raw unchanged: archive (1)/order.csv | PASS | 1a166b6b9daa4b31 |
| Raw data | raw unchanged: archive (1)/product.csv | PASS | 58f04efce65d724d |
| Raw data | raw unchanged: supply_chain_dataset1.csv | PASS | 76dad280541a60f1 |
| Raw data | raw CSVs identical to git HEAD | PASS | git diff --quiet HEAD -- raw CSVs |
| Integrity | profile pairs invented (not in source) | PASS | 0 |
| Integrity | source pairs missing from profile | PASS | 0 |
| Integrity | profile pairs == lane pairs | PASS | 211 |
| Integrity | profile duplicate (sku, supplier) | PASS | 0 |
| Integrity | lanes duplicate (sku, warehouse) | PASS | 0 |
| Integrity | profile nulls | PASS | 0 |
| Integrity | lanes nulls | PASS | 0 |
| Integrity | available_qty within historical range 200-499 | PASS | [200, 499] |
| Integrity | defect_rate within [0, real max] | PASS | [0.0118, 0.1442, 0.144299] |
| Integrity | fulfillment_rate within [0, 1] | PASS | [0.7518, 0.8686] |
| Integrity | quality fields constant per supplier | PASS | 10 suppliers |
| Integrity | SQLite integrity_check | PASS | ok |
| Integrity | schema user_version | PASS | 2 |
| Reproducibility | two in-memory generations identical | PASS | seed 20260926 |
| Reproducibility | file deleted then regenerated byte-identical | PASS | {"before": "3ef2933fa9488cc1", "after": "3ef2933fa9488cc1"} |
| Reproducibility | regenerated output hash matches metadata | PASS | 3ef2933fa9488cc1 |
| Future | time-series columns for demand ML retained | PASS | ["date", "demand_forecast", "inventory_level", "order_quantity", "promotion_flag", "region", "reorder_point", "sku_id", "stockout_flag", "units_sold", "warehouse_id"] |
| Future | 365 daily observations per SKU-warehouse retained | PASS | 365 |
| Future | raw criteria on every candidate | PASS | ["available_qty", "defect_rate", "fulfillment_rate", "heuristic_score", "lane_count", "lead_time_days", "lead_time_max", "lead_time_min", "rank", "raw_score", "scoring", "sku_id", "supplier_id", "unit_cost", "unit_cost_max", "unit_cost_min", "warehouses"] |
| Future | per-supplier availability retained (split orders) | PASS | 211 |

## Row-count validation (raw source vs database)

| Source | Table | Source rows | DB rows | Difference |
|---|---|---|---|---|
| DataSets/archive (1)/product.csv | products | 1000 | 1000 | 0 |
| DataSets/archive (1)/order.csv | orders | 1000 | 1000 | 0 |
| DataSets/archive (1)/customer.csv | customers | 1000 | 1000 | 0 |
| DataSets/archive (1)/clander.csv | calendar | 1000 | 935 | 65 |
| DataSets/Procurement KPI Analysis Dataset.csv | procurement_kpi | 777 | 777 | 0 |
| DataSets/supply_chain_dataset1.csv | supply_chain | 91250 | 91250 | 0 |
| DataSets/simulated/supplier_sku_profile.csv | supplier_sku_profile | 211 | 211 | 0 |
| supply_chain (distinct SKU × warehouse) | sku_supplier_lanes | 250 | 250 | 0 |

The only difference is the calendar's 65 duplicate dates, removed by the pre-existing loader.

## End-to-end procurement test (every urgent SKU)

| SKU | Urgent WH | Need | Eligible | Excluded (Rule 5) | Chosen | Status | Backup | Plan |
|---|---|---|---|---|---|---|---|---|
| SKU_44 | WH_2, WH_3 | 790 | 0/5 | SUP_9, SUP_5, SUP_2, SUP_7, SUP_1 | none | - | - | no eligible supplier |
| SKU_19 | WH_3, WH_4 | 720 | 0/4 | SUP_7, SUP_10, SUP_8, SUP_1 | none | - | - | no eligible supplier |
| SKU_37 | WH_1, WH_5 | 610 | 0/4 | SUP_4, SUP_10, SUP_7, SUP_8 | none | - | - | no eligible supplier |
| SKU_2 | WH_3 | 410 | 1/5 | SUP_5, SUP_2, SUP_10, SUP_7 | SUP_9 | Standard | - | planned |
| SKU_47 | WH_4 | 390 | 0/4 | SUP_1, SUP_5, SUP_10, SUP_7 | none | - | - | no eligible supplier |
| SKU_42 | WH_1 | 390 | 1/5 | SUP_7, SUP_2, SUP_3, SUP_5 | SUP_4 | Standard | - | planned |
| SKU_41 | WH_1 | 370 | 2/3 | SUP_3 | SUP_6 | Standard | SUP_1 | planned |
| SKU_39 | WH_1 | 370 | 1/4 | SUP_6, SUP_4, SUP_10 | SUP_3 | High Risk | - | planned |
| SKU_50 | WH_5 | 360 | 1/4 | SUP_10, SUP_8, SUP_2 | SUP_4 | Standard | - | planned |
| SKU_21 | WH_3 | 340 | 1/3 | SUP_1, SUP_8 | SUP_2 | High Risk | - | planned |
| SKU_5 | WH_4 | 300 | 2/3 | SUP_1 | SUP_7 | High Risk | SUP_8 | planned |

- Different SKUs → different rankings: 50 distinct rankings across 50 SKUs; top supplier counts {'SUP_1': 5, 'SUP_2': 5, 'SUP_3': 4, 'SUP_4': 4, 'SUP_5': 10, 'SUP_6': 3, 'SUP_7': 7, 'SUP_8': 1, 'SUP_9': 7, 'SUP_10': 4}.
- Insufficient availability excluded: every ranked candidate has available ≥ need and every excluded one < need (tested for all 11 urgent SKUs).
- No eligible supplier: SKU_44, SKU_19, SKU_37, SKU_47 → no recommendation, no plan, combined availability reported.
- Non-urgent SKU_3 → plan status 'no_action' (no purchase order).
- Before: 53 of 53 urgent products → SUP_1 / SUP_9 (old code re-run on backup DB).
- API tests: SKU pipeline, 410 for product_id, 404 for unknown SKU, 422 for invented pair or excessive quantity, PO creation with sku_id.

# 18. Reproducibility

| Item | Value |
|---|---|
| Simulation seed | 20260926 |
| Random streams | numpy.random.SeedSequence(seed).spawn(3) -> [available_qty, defect_rate, fulfillment_rate] |
| Generation script | backend/simulation/supplier_sku_profile.py |
| Simulated data generated | 2026-09-27T17:08:02 |
| Migration date | 2026-09-27 |
| Output SHA-256 | 3ef2933fa9488cc1b544dea678eb8c7560dd142db30f80116fd68caacff866ab |
| Input SHA-256: DataSets/supply_chain_dataset1.csv | 76dad280541a60f135442f3eeb3ef4804a28e48f6a9677f63ddba236ddc80491 |
| Input SHA-256: DataSets/Procurement KPI Analysis Dataset.csv | bf9a529c52cd7de254994a55d087f865d3aefc0ba66790e8ee43a26e7beb6e9c |
| Python (report build) | 3.13.5 (win32); app pinned to 3.11 for deployment |
| Backup | data/backup/pre_migration_2026-09-27/ (raw CSVs + pre-migration DB, verified by checksum) |

## Commands

| Purpose | Command |
|---|---|
| Verify committed simulated CSV | python -m backend.simulation.supplier_sku_profile --check |
| Regenerate simulated dataset | python -m backend.simulation.supplier_sku_profile |
| Rebuild database | python backend/database.py (or: python run_app.py --rebuild-db) |
| Run tests | python -m pytest backend/tests |
| Capture state snapshot | python scripts/migration/capture_state.py <before\|after> |
| Validate migration | python scripts/migration/validate.py |
| Regenerate change / deletion logs | python scripts/migration/change_log.py |
| Regenerate this report | pip install reportlab && python scripts/migration/build_report.py |
| Verify raw files | cd DataSets && sha256sum -c CHECKSUMS.sha256 |

Regeneration was tested by deleting the simulated CSV and regenerating it: the new file is byte-identical (Section 17, Reproducibility checks).